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REVIEW ARTICLE

Flood resilience in cities and urban agglomerations: a systematic review of hazard causes, assessment frameworks, and recovery strategies based on LLM tools

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Abstract
With the increasing frequency of extreme weather events, the study of flood resilience in cities and urban agglomerations has become a critical topic for addressing climate change. This paper systematically reviews three key research directions—flood causes, assessment frameworks, and resilience enhancement strategies—by integrating quantitative and qualitative approaches across 412 relevant studies. The study innovatively employs large language models (LLMs) and machine learning techniques to achieve a structured research synthesis. The study identifies four major issues in current research on flood resilience in cities and urban agglomerations: (1) Existing studies predominantly focus on static indicators for individual cities, neglecting the dynamic interactions within urban agglomeration systems, making it challenging to reveal the real characteristics of cross-regional resource allocation and disaster propagation. (2) Governance mechanisms in urban agglomerations lack operational feasibility; conflicting interests among cities often reduce regional synergy to mere formalities. (3) There is an overemphasis on real-time risk monitoring and assessment models, while the practical value of disaster prediction models and pre-disaster planning is often overlooked. While existing studies consider inequities in regional and population vulnerabilities, they lack focus on integrating these needs into resilience frameworks and their practical implementation. Based on these findings, this study recommends incorporating the compounding effects of multidimensional factors, enhancing the implementability of regional collaborative governance and pre-disaster planning, and embedding economic practicality and universal applicability into the design of resilience assessment and enhancement frameworks.

Keywords Urban resilience · Flooding · Urban agglomerations · Large language models · Active machine learning

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1 Introduction

In recent years, global warming has intensified typhoons and torrential rains, while rapid urbanization has led to inefficient land use practices and fragile flood protection infrastructure. This has resulted in an increased frequency and severity of flooding, causing approximately $700 billion in economic losses worldwide over the past 20 years (He et al. 2024). Existing studies show that engineering-based disaster prevention is a traditional and effective measure against floods (Boulange et al. 2021). However, relying solely on upgrading engineering standards to strengthen systemic resistance is economically unsustainable for addressing increasingly extreme flood events. Moreover, the evaluation of Urban Flood Resilience (UFR) encompasses multidimensional aspects—social, economic, ecological, and managerial—beyond engineering dimensions (Li et al. 2023). Consequently, recent flood prevention paradigms have shifted from resisting nature to adapting to it, adopting a systematic disaster prevention concept that couples the multidimensional vulnerabilities of disaster-bearing spaces (Zhu et al., 2024a). In disaster prevention, resilience refers to the ability of cities to maintain dynamic balance in internal structures and functions when subjected to external disturbances (Folke 2006). Focusing on flood management, resilience emphasizes the construction of multifunctional flood-bearing systems, robust disaster mitigation facilities, and highly connected sponge ecological networks (Wang et al. 2024a, b, c, d, e, f).

As the primary spatial carriers of socio-economic activities and population aggregation in modern society, cities exhibit significant complexity in flood response (Zhao et al. 2024a, b). At the urban scale, the disaster mechanism of flooding often manifests as basin floods entering urban cores through river networks, superimposed with localized heavy rainfall, resulting in steep flood peaks and overloaded drainage systems (Yuan et al. 2024). The structural characteristics of urban disaster-bearing spaces, such as high-density building clusters and complex underground spaces, make flooding particularly damaging to critical functional areas like central business districts and transportation hubs (Deopa et al. 2024). Urban agglomerations, as the predominant form of urbanization in the context of regional collaborative development, represent large systems characterized by complementary regional infrastructure, economic markets, and social management systems (Fang and Yu 2017). On one hand, the disaster mechanism of flooding at mega-spatial scales is highly complex, often involving multi-hazard coupled reactions within intricate disaster chains (Gill and Malamud 2017). For instance, heavy precipitation from a strong typhoon may cause massive basin flooding and geological disasters within an urban agglomeration, leading to the collapse of hydrological system capacity and infrastructure failure in affected cities (Wang et al. 2024a, b, c, d, e, f). On the other hand, the physical disaster-bearing spaces of urban agglomerations are structurally and characteristically more complex than those of individual cities. This complexity is reflected in the strong clustering effect of population distribution, significant regional variability in flood protection infrastructure development (Ji et al. 2024), and considerable differences in disaster response capacities among management and supply systems within different cities of the agglomeration (Pollack et al. 2024). Consequently, flood resistance in the physical spatial dimension of urban agglomerations exhibits substantial regional inequities (Cano Pecharroman and Hahn 2024). Given the complex causes of flooding and the characteristics of disaster-bearing physical spaces, the severity and risk of flooding at the scale of urban agglomerations are significantly greater than those in individual cities.

Existing research on flood resilience can be broadly categorized into disaster risk assessment and systemic resilience enhancement strategies. Disaster risk assessment facilitates the quantification of urban resilience, with foundational studies focusing on integrating social, economic, and environmental indicators into comprehensive resilience evaluation models to quantify regional flood risks. Common methodologies include system dynamics, multi-criteria decision-making (MCDM), fuzzy logic, and network analysis (Meerow and Newell 2016; Song et al. 2019). At the urban scale, the indicators are typically associated with factors such as urban infrastructure, building density, local topography, and response speed. These assessments are characterized by high data accuracy and relatively low difficulty in integrated analysis, with a primary focus on intra-urban flood risk management, infrastructure resilience, and emergency response capacity (Pollack et al. 2024). At the urban agglomeration scale, indicators should focus more on intercity interactions, regional hydrological cycles, and broader land use and coverage changes (Ji et al. 2024). However, challenges arise due to broader data coverage requirements and resolution inconsistencies, as well as the lack of objective, quantitative methods for evaluating intercity connectivity. Consequently, most studies conduct risk assessments at the city level, generating regional maps of urban resilience level differences (Liao et al. 2024; Zhang et al. 2024). Fundamentally, these are evaluations of urban resilience rather than resilience at the urban agglomeration scale. Therefore, there is a need to explore more scientific and precise assessment methods on a macro scale.

Resilience enhancement strategies are a follow-up to the quantitative assessment of flood risks. Most strategy studies continue to focus on enhancing resilience at the urban scale through engineering and ecological systems (D'Ambrosio and Longobardi 2023; Liu et al. 2023), with limited innovation in the dimension of regional collaborative governance. Current research has proposed comprehensive macro-scale disaster-adaptive construction methods based on flood resilience characteristics. However, these are mostly organized around watersheds as the primary unit (Najafi et al. 2024; Wang et al. 2023a, b, c, d, e), lacking deep control methods for managing floods from watersheds to urban agglomerations. Furthermore, many of these solutions lack systematic practical strategies integrating urban planning and management. Resilience enhancement strategies at the urban agglomeration scale could bridge the gap between watershed and city scales, enabling three-tier collaborative flood management across watersheds, urban agglomerations, and cities. In addition, there is limited dedicated research on the causes of flooding in urban agglomerations. Theoretical studies on the mechanisms of disaster evolution across the full temporal cycle are effective for revealing the coupling dynamics between urban agglomeration systems and floods. Such research can help mitigate impacts across all stages of disasters, from initiation and development to evolution and recession.

Existing flood resilience review methods primarily rely on qualitative selection and bibliometric analysis, yet they exhibit certain limitations in studying flood resilience across complex, multi-scale contexts. Qualitative reviews depend on researchers' subjective selection of literature and thematic synthesis, making it challenging to systematically extract key flood resilience factors across large-scale, interdisciplinary studies (Long et al. 2024; Prashar et al. 2024). Bibliometric tools, such as CiteSpace and VOSviewer, primarily rely on keyword co-occurrence and citation networks. While they effectively identify research trends, they struggle to capture the core research logic embedded within unstructured texts and fail to quantify the relationships between different research trajectories. Consequently, these methods remain constrained in addressing the multidimensional interactions, cross-scale adaptability, and systemic integration required for comprehensive flood resilience research (Li et al. 2025; Shah and Sharifi 2025). In recent years, LLM-based interactive analytical frameworks have been widely applied in academic literature reviews across various fields. Their primary advantage lies in the ability to automatically process large-scale unstructured texts and enhance information integration efficiency through structured synthesis (Delgado-Chaves et al. 2025; Scherbakov et al. 2025). Compared to traditional review methods, LLM offers several distinct advantages: (a) Minimizing the influence of subjective selection, thereby enhancing the objectivity of literature reviews; (b) Incorporating contextual information to conduct in-depth analyses of research trajectories, overcoming the limitations of traditional bibliometric tools that are restricted to keyword-based analysis; (c) Quantifying research patterns within complex multi-scale systems, making it particularly suitable for interdisciplinary and multi-variable flood resilience analysis. Therefore, the LLM approach effectively addresses the shortcomings of traditional review methods in flood resilience research.

Building on this, this study employs ASReview in conjunction with an LLM API-based interactive analytical framework to optimize literature selection, analyze research trajectories, and systematically synthesize studies on urban and urban agglomeration flood resilience. The objectives of this study are as follows: (a) To identify the specific causes of flood disasters at both urban and urban agglomeration scales; (b) To summarize the key factors and mechanisms influencing flood risk and examine their contradictions and synergies; (c) To synthesize strategies for enhancing flood resilience in cities and urban agglomerations and propose optimization recommendations.

2 Method

This study adopts methods of data collection, screening, information extraction, and analysis to explore the differences and gaps in flood resilience research between cities and urban agglomerations (Fig. 1). The research methodology comprises four main stages: data collection and screening using the ASReview active learning algorithm, information extraction and processing with the assistance of Large Language Models (LLMs), data analysis, and the presentation and discussion of results. To provide a clearer visualization of the overall methodological framework, we have created a detailed workflow diagram (Fig. 2) to assist readers in understanding the logical relationships and scientific basis of each step.

2.1 Document screening process

In the third week of October 2024, two rounds of literature searches were conducted in the Web of Science Core Collection to identify studies closely related to the research topic. Given that flood resilience evaluation research has gained widespread attention in recent years, and studies incorporating big data and machine learning in this field have only recently become prevalent, the search was limited to publications from 2021 to 2024 to ensure methodological comparability. The first round of searches employed keywords such as “urban agglomeration,” “Megalopolis,” “urban cluster,” and “city agglomeration” in combination with “flood” and “rainstorm,” focusing on studies related to flood resilience in urban agglomerations. This yielded 22 articles. The second round included broader keywords such as “urban,” “city,” “flood,” “rainstorm,” and “resilience” to capture a wider range of studies on urban flood resilience, resulting in 486 articles. During the initial screening process, all duplicate papers were removed

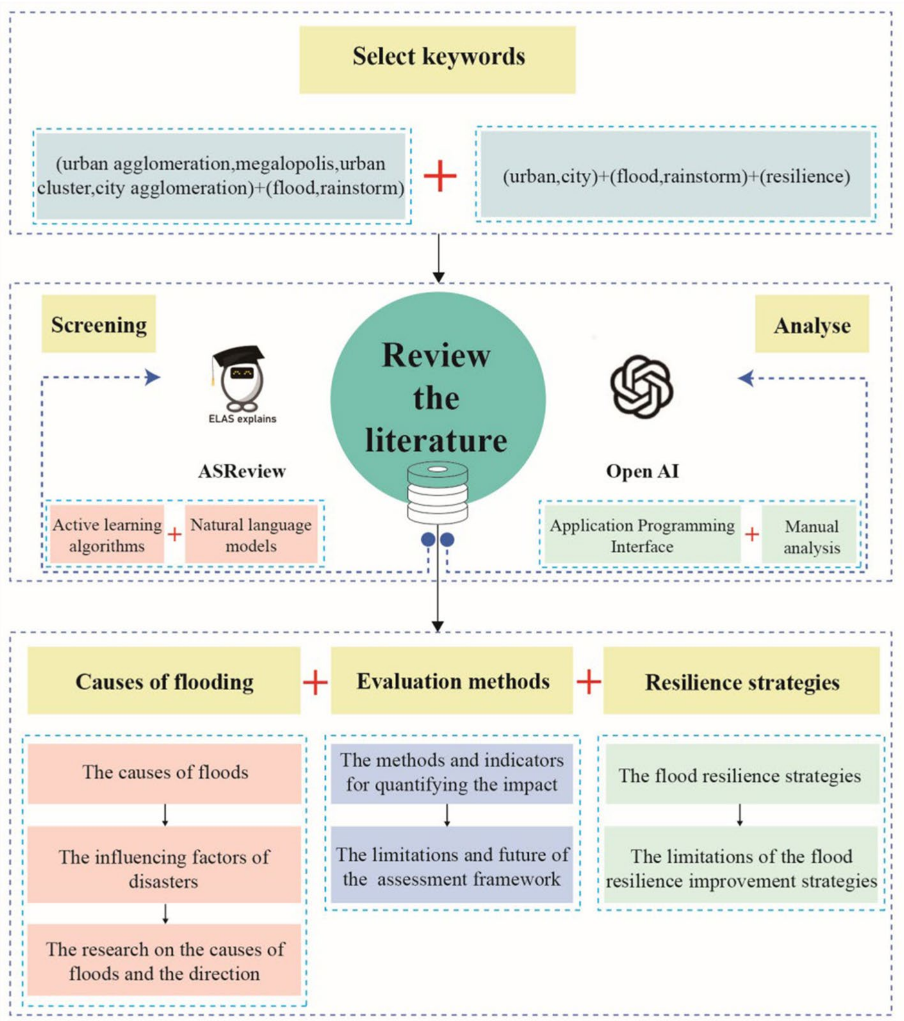

Fig. 1 Research framework of this review

to ensure the uniqueness of the final dataset. This study employed DOI matching and manual cross-checking to enhance the rigor of the screening process. After confirming that there was no overlap between the two rounds of searches, a total of 508 unique articles were included (denoted as N1 in Fig. 2). The purpose of the first round was to focus on the unique aspects of urban agglomeration resilience in flood management, while the second round adopted a broader perspective to examine overall urban flood resilience. This step-by-step screening strategy ensured that we could focus on the distinctiveness of urban agglomerations while also establishing comparisons through a broad range of urban studies to identify commonalities and differences, thereby providing comprehensive data to support comparative research.

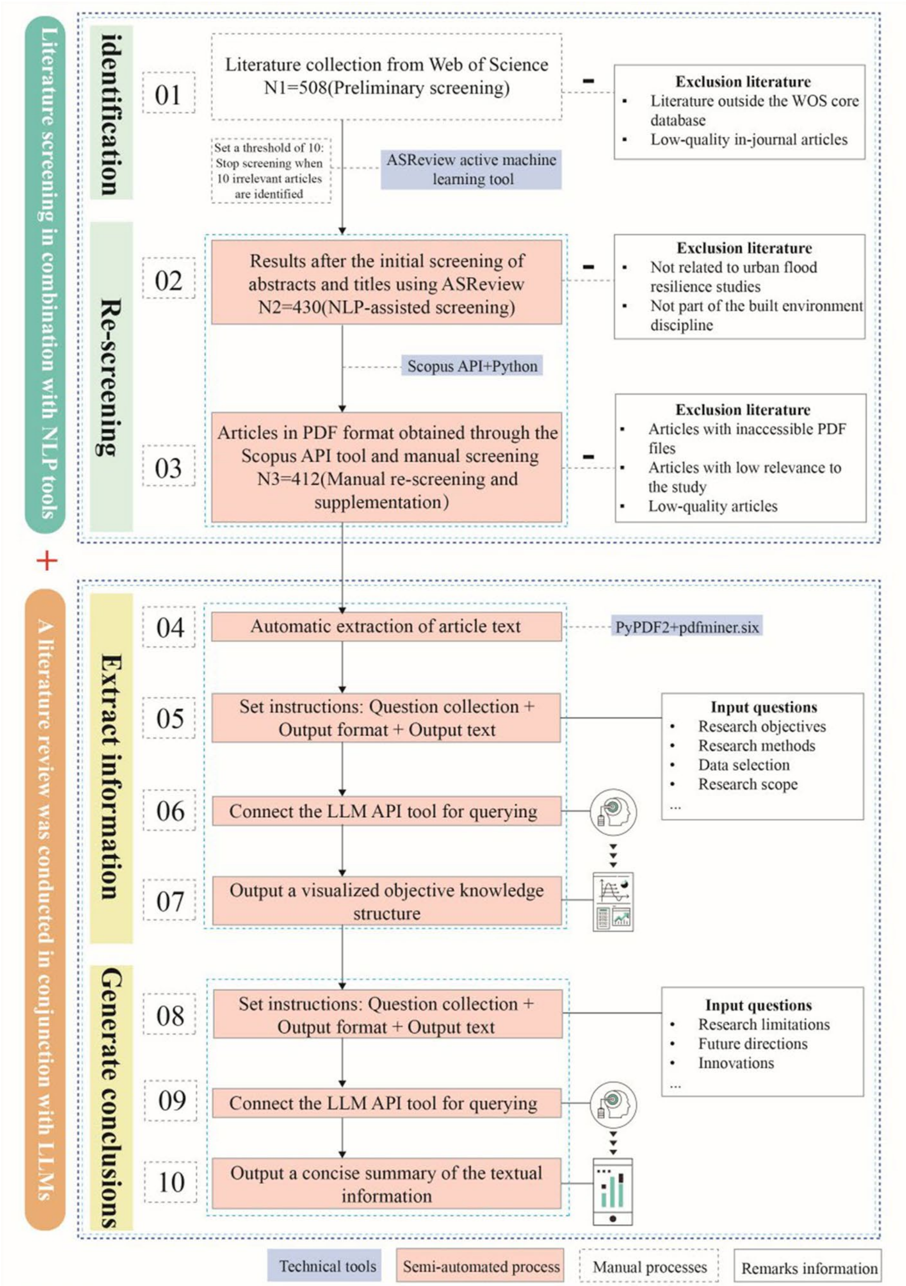

Fig. 2 Workflow diagram for literature screening using active learning tools and semi-automated analysis incorporating LLM

2.2 Using active machine learning algorithms for topic screening

After initially identifying 508 articles, we used ASReview, an open-source machine learning tool developed by Rens van de Schoot and colleagues, to perform topic screening. This tool is designed to assist researchers in the literature screening process for systematic reviews and meta-analyses (van de Schoot et al. 2021). ASReview integrates state-of-the-art active learning algorithms to optimize the screening process through human–computer interaction, enabling the intelligent and efficient identification of articles highly relevant to the research topic. Compared to traditional manual screening methods, ASReview offers significant advantages in saving time and reducing labor input.

Specifically, ASReview employs an uncertainty sampling strategy to analyze the titles and abstracts of articles, prioritizing those with uncertain relevance for manual evaluation. This ensures that each round of screening effectively improves the quality of the selection process. In this study, we set a threshold to terminate screening when ten consecutive irrelevant articles were identified, ensuring both efficiency and precision in the process. We selected the Random Forest model as the foundational screening algorithm for this study due to its excellent performance and adaptability in handling literature classification tasks (Pedregosa, 2011). In addition, the screening process incorporated a Natural Language Processing (NLP) model to further analyze the topics and content of the articles. By conducting in-depth analyses of key terms, research objectives, and methods, the NLP model effectively identified articles most relevant to the research topic. During the screening process, some articles were excluded due to their irrelevance to the research topic. For instance, some studies focused on the quantitative assessment of socio-economic impacts without addressing urban resilience construction. Other articles focused on rural or non-urban areas, with findings that had low relevance to urban flood resilience construction, and were therefore not included in the subsequent analysis. Ultimately, we selected 430 articles (denoted as N2 in Fig. 2) that were highly relevant to the research topic for subsequent analysis.

2.3 Text recognition and extraction based on PyPDF2 and pdfminer.six

This study employs Large Language Models (LLMs), specifically ChatGPT-4O, in combination with traditional review methods to achieve cognition of macro-knowledge structures. With a deep architecture of 175 billion parameters, ChatGPT-4O surpasses traditional models (e.g., BERT with 340 million parameters) in contextual understanding and reasoning for complex literature analysis. It can accurately extract multi-level information and outperforms similar models in public testing, achieving over 90% accuracy on multi-task benchmarks (Andrea Sottana, 2023). The preliminary work involved obtaining full-text PDFs of all relevant literature. This study utilized Python toolkits PyPDF2 and pdfminer.six to extract key textual content such as research objectives, methods, and conclusions. During the information extraction phase, the Scopus API was used to download full-text articles from certain Elsevier journals. For articles not accessible via the API, manual searches were conducted to supplement the dataset. Ultimately, 412 complete PDF files were collected (denoted as N3 in Fig. 2), providing reliable data support for the subsequent two rounds of systematic questioning with GPT-4O.

2.4 LLM-based knowledge extraction and insight identification

After obtaining the PDFs and extracting text from the literature, this study utilized LLMs for knowledge extraction and systematic analysis. The main approach in this phase involved employing Large Language Models (LLMs), specifically GPT-4O, to conduct two rounds of systematic questioning on the literature (complete code in Appendix Fig. 10). This enabled a progressive understanding from macro-knowledge structures to specific domains, exploring the commonalities and differences in flood resilience research between cities and urban agglomerations, while identifying gaps and future research opportunities.

The purpose of the first round of questioning (Step 6 in Fig. 2) was to establish a knowledge framework for the literature. First, a Python-based script was developed to extract text segments, followed by systematic questioning of 412 articles using GPT to train the model for systematic comprehension of the literature. To ensure comprehensive coverage of key topics in flood resilience research, this round of questioning was structured around core dimensions such as research objectives, methodologies, study content, and conclusions. A structured question design was employed to ensure the systematic and comprehensive extraction of information. The study optimized the question classification logic to enable the LLM to accurately distinguish between similar concepts, reduce ambiguity, and adapt to different spatial scales and methodological approaches. The complete set of first-round questions and corresponding JSON code is provided in the appendix to support research reproducibility and further validation.

To ensure scientific accuracy in answer matching, the code in this study optimized classification logic using vectorization methods, enabling the LLM to correctly classify similar descriptions with varying wordings. Additionally, to enhance efficiency and avoid the negative impact of irrelevant fields, preprocessing code was designed to exclude unrelated sections such as references. Progressive questioning logic was implemented, prioritizing queries from abstracts and introductions, and extending to full documents if answers to specific questions were found insufficient.For cases where a single article might provide multiple answers to a question, a weighting mechanism was implemented in the code (complete code in Appendix Fig. 11). This included generating primary classification statistics and comprehensive classification results, ultimately reflecting the scientific distribution of answers through normalized statistical outcomes. Subsequently, to verify the accuracy of the questioning results, 5% of the 412 articles (approximately 21) were randomly sampled, and GPT-4O's generated answers were compared with the original literature content on a case-by-case basis. The results indicated that GPT-4O achieved approximately 96% accuracy in extracting research objectives, methodologies, and conclusions, demonstrating the high reliability of the model's application in this study. This study summarized the framework and considerations for structured questioning (Fig. 3) as a reference for similar research.

Building upon the knowledge framework established in the first round of questioning, the second round of questioning (Step 9 in Fig. 2) further analyzed differences among studies and explored future research trends. This round of questioning focused on key research directions at both the urban and urban agglomeration scales, including the applicability of flood impact assessment methods, the effectiveness of resilience enhancement strategies, comparisons of governance models across different regions, and emerging research trends. Furthermore, to enhance the scientific validity of LLM-derived conclusions in complex research contexts, this study employed a weighted

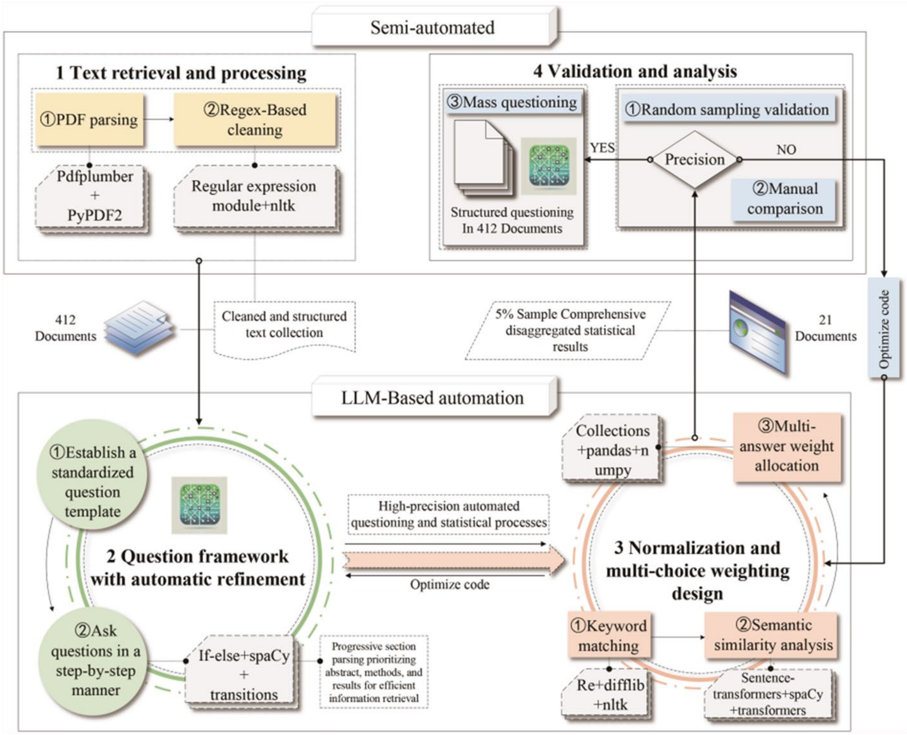

Fig. 3 Overview of structured questioning code mechanisms and toolkits

allocation mechanism in the second round of questioning to ensure appropriate prioritization of different issues and to facilitate reasonable forecasting of research trends. The complete set of second-round questions is provided in the appendix Fig. 12.

3 Overall review

This chapter presents systematic findings from the structured questioning of 412 articles. Current systematic research on flood resilience is predominantly based on three processes: (1) Understanding the interaction between disaster-bearing spaces and hazard agents to identify the causes of disasters; (2) Quantifying the impacts of floods and secondary hazards within disaster chains and summarizing the spatiotemporal evolution of flood resilience; (3) Proposing context-specific strategies for enhancing flood resilience, typically covering fields such as engineering, management, economics, ecology, and urban planning. The study integrates eight related questions and answers into a dendrogram (Fig. 4): (1) Research scale; (2) Research type; (3) Analytical tools; (4) Data types; (5) Research directions; (6) Focus on flood causes; (7) Quantitative methods for flood impacts; (8) Strategies for enhancing flood resilience.

The analysis reveals three main characteristics of current flood resilience research. First, there is a centralization of research scales. The urban scale is the most prominent unit, accounting for approximately 57.8%, while the urban agglomeration scale accounts for

Fig. 4 Dendrogram collection of structured questioning conclusions for 412 articles using LLM API tools

24.3%. Other scales, such as watershed, cross-scale, and county scales, are significantly underrepresented. This reflects a lack of research on cross-scale interactions, limiting the understanding of systemic coupling between cities and surrounding regions in flood risk propagation and control. Second, the integration of research methods is insufficient. Quantitative studies dominate, comprising 77.9%, while qualitative studies and mixed methods account for 14.1 and 8%, respectively. The dominance of quantitative approaches stems from the widespread use of data-driven and indicator systems, which effectively quantify flood resilience performance and dynamic evolution. However, reliance solely on quantitative data analysis may overlook deeper social, institutional, and ecological impacts (Aroca-Jiménez et al. 2022). Meanwhile, the application of advanced techniques such as machine learning and network analysis is underutilized, accounting for only 8.8 and 9.2%, respectively, indicating that current research has not fully harnessed the potential of big data tools in resilience analysis. Third, there is a coexistence of data diversity and limitations. Environmental data (24.6%) and social resilience and emergency response data (19.5%) are the primary focus areas for data application, while imaging data accounts for only 1.3%. This reflects insufficient attention to real-time evaluation using remote sensing technologies in current research.

In the specific research area of flood causes, the focus is on the vulnerability of disaster-bearing systems and the risk spillover from human interference, accounting for 42.2 and 41.1%, respectively. These results indicate that current research pays more attention to internal system vulnerabilities and the amplifying effects of external human activities on flood risk. However, studies on disaster chain effects are relatively few, accounting for only 16.4%, indicating insufficient understanding of the systemic evolution of flood risk. In terms of resilience enhancement strategies, management measures (35.9%) and improvements in the physical resilience of disaster-bearing systems (32.7%) are the main approaches. However, ecological restoration strategies account for only 9.1%, highlighting the need for greater attention to sustainable ecological restoration strategies (Jeong et al. 2021). Comprehensive governance strategies account for 22.3%, reflecting preliminary exploration of cross-sectoral and cross-regional collaborative governance mechanisms. In the research area of quantitative flood impact assessment, the indicator scoring method has the highest share at 45%, followed by quantification of affected area (22.2%) and environmental impact assessment (19.3%). However, economic loss estimation and casualty statistics account for only 9.1 and 3.5%, respectively, indicating insufficient emphasis on socio-economic dimensions in the indicator system and neglect of the population vulnerability perspective (Yang et al. 2024).

4 Comparative analysis

This chapter adopts a framework of flood causes, risk assessment, and resilience enhancement strategies to uncover the complex interaction mechanisms between the built environment and hazard entities at different scales, as well as the research logic of multi-scale governance. Using the keyword system presented in the word cloud (Fig. 5), the chapter outlines the intrinsic connections and distinctions from the disaster characteristics of physical spaces to regional disaster chains, while emphasizing the core pathways for flood resilience assessment and enhancement at different scales. The three main research directions encompass the following topics at the city and urban agglomeration scales: (1) Causes of urban flooding; (2) Causes of urban agglomeration flooding; (3) Quantification and assessment of urban flood impacts; (4) Quantification and assessment of urban agglomeration flood impacts; (5) Strategies for enhancing urban flood resilience; (6) Strategies for enhancing urban agglomeration flood resilience. The subsequent sections of this chapter will provide a detailed analysis of these topics.

4.1 Analysis of flood causes

4.1.1 Urban scale: local responses to macro-disaster chains

The causes of urban flooding reflect both the vulnerability of the urban built environment and the influence of macro-disaster chains. Their multi-level interactions jointly determine the mechanisms and characteristics of urban flood formation (Fig. 5, top row, left).

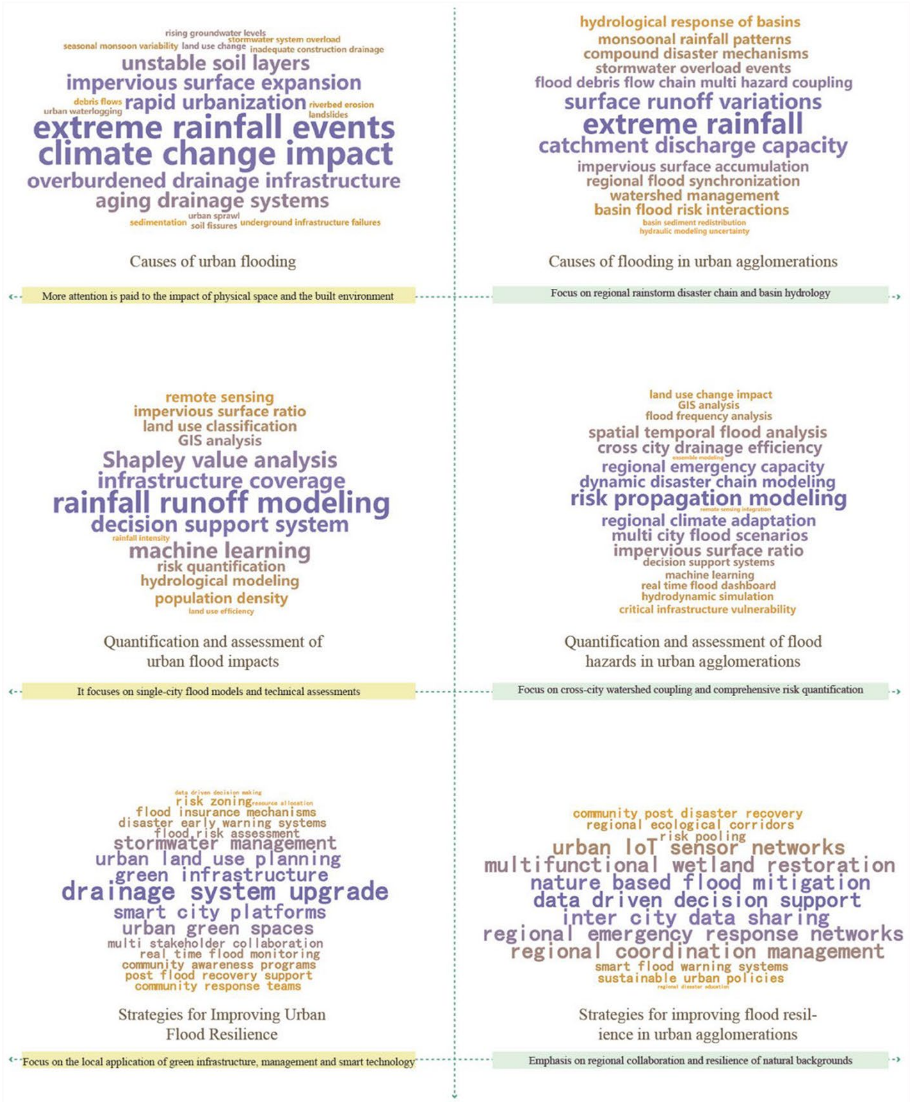

Fig. 5 Word cloud collection of the three main research directions on flood resilience in cities and urban agglomerations

From the perspective of the built environment, rapid urbanization has significantly altered urban surface characteristics. The rapid increase in impervious surface coverage inhibits natural rainwater infiltration, leading to a surge in surface runoff. The inability of drainage systems to efficiently divert this runoff has become one of the core reasons for frequent urban flooding (Bernardini et al. 2024; Borowska-Stefańska et al. 2024). Additionally, urban expansion often neglects the development of green infrastructure, which not only weakens the city's ability to regulate hydrological processes but also amplifies the clustering effects of flooding in high-density building areas.

From the perspective of the coupling relationship between disasters and cities, the frequency and intensity of extreme rainfall events are direct triggers. High-intensity rainfall over short periods overwhelms urban drainage systems, with low-lying areas and regions with aging infrastructure being particularly vulnerable (Jiang and Tan 2022; Luo and Zhang 2022; Nelson and Ahmadpoor 2023). Additionally, processes such as regional rainfall distribution and watershed sediment transport affect the speed and extent of flood propagation. This upstream-to-downstream linkage creates significant superimposed amplification effects within urban spaces (Wang et al. 2022; Wang et al. 2023a, b, c, d, e).

From the perspective of disaster chains, urban flood formation can be seen as a localized response within a regional disaster chain (Qiao Wang, Ruijia Zhang, et al. 2024). This chain involves meteorological drivers, watershed runoff, and multiple interactions within downstream urban spaces. Extreme rainfall, as a driving force, not only directly overloads urban drainage systems but also transmits flood pressures from upstream to urban interiors through regional hydrological processes, intensifying the local concentration and diffusion of disasters (Rajput et al. 2023).

Different types of cities exhibit distinct flood causes, primarily influenced by geographical location, precipitation patterns, urban morphology, and infrastructure development levels. Coastal cities (e.g., Shanghai, New York, Guangzhou) are affected by the combined impact of typhoon-induced heavy rainfall and storm surges, making them prone to extreme floods caused by tidal backflow and short-term intense precipitation(Deng et al. 2016). High-density cities (e.g., Tokyo, London, Singapore) experience limited rainwater infiltration due to highly impervious surfaces, increasing drainage pressure and making localized heavy rainfall more likely to cause urban flooding (Sado-Inamura and Fukushi 2019). Inland low-lying cities (e.g., Wuhan, Chicago, Beijing) are mainly affected by low-lying terrain and short-duration intense rainfall, leading to difficulties in timely flood discharge and localized waterlogging due to overloaded drainage systems (Zhao et al. 2019). Developing cities (e.g., Mumbai, Jakarta, Lagos) often suffer from prolonged urban flooding and greater socio-economic losses due to inadequate drainage infrastructure and unregulated urban expansion, making them particularly vulnerable to extreme precipitation events (Zope et al. 2014). Overall, while different cities share common flood-inducing factors such as increased precipitation intensity and rising drainage pressure, their specific manifestations are shaped by urban characteristics, leading to significant regional disparities.

4.1.2 Urban agglomeration scale: multi-scale and full-chain risk propagation

Compared to the urban scale, research on the causes of urban agglomeration flooding places greater emphasis on cross-regional characteristics (Fig. 5, top row, right). Building on the climatic dynamics and urbanization characteristics of individual cities, urban agglomerations exhibit more complex features in terms of spatial connectivity, disaster chain propagation, and regional hydrological processes (Wei et al. 2023).

The cascading effects of watershed flooding not only amplify the runoff caused by impervious surface expansion and concentrated building layouts from urbanization but also strengthen the networked propagation pathways of floods at the urban agglomeration scale (Zhang et al. 2024). The disruption of ecological connectivity reduces the coherence of regional hydrological cycles, thereby weakening the ability of natural buffering systems to regulate floods within the region (Chen et al. 2024a, b, c, d). The disaster chain characteristics of urban agglomerations reflect the intricate interactions between complex climatic systems and regional geographical patterns, with intensity and complexity far exceeding those of individual cities. For example, climatic drivers such as the East Asian monsoon and the El Niño phenomenon enhance the synchronicity and concentration of regional precipitation, exacerbating the transmission effects of disaster chains and forming a regional diffusion mechanism where upstream precipitation pressures propagate to downstream cities (Doan et al. 2022). When disaster chains transition to urban areas, high-intensity watershed runoff accelerates the spread of floods between cities and intensifies the cumulative impacts of insufficient upstream water retention on downstream cities.

Furthermore, the agglomeration effects of socio-economic factors exacerbate the compounded exposure and vulnerability of urban agglomerations. High-density economic activities and population distributions increase disaster risks to critical infrastructure and residential areas, while imbalances in urban–rural development further widen disparities in flood resistance capabilities among populations. Key drivers of flood exposure, such as population density, GDP, and land use intensity, highlight the resilience disparities within urban agglomerations (Liu et al. 2022).

Different types of urban agglomerations exhibit significant variations in flood causes, primarily influenced by hydrological terrain, precipitation patterns, and urban spatial structures. Delta urban agglomerations (e.g., the Pearl River Delta, Yangtze River Delta, Randstad in the Netherlands) are affected by the combined influence of tidal forces, river networks, and typhoon-induced heavy rainfall, with flooding primarily occurring due to storm surge backflow, rising river water levels, and extreme precipitation coupling effects. For example, the Pearl River Delta frequently experiences concurrent heavy rainfall and tidal backflow, leading to severe urban waterlogging in cities such as Guangzhou and Shenzhen (Cheng and Chen 2017). In contrast, basin urban agglomerations (e.g., Chengdu Plain, Mexico Valley, California's Central Valley) face challenges due to enclosed topography and drainage constraints, making it difficult for floodwaters to discharge, often resulting in prolonged water accumulation in low-lying areas(Payne et al. 2020). Additionally, typhoon-affected urban agglomerations (e.g., Beijing-Tianjin-Hebei, the Min Delta, the Kanto region in Japan) are primarily impacted by short-duration extreme rainfall events triggered by typhoons, which significantly increase regional hydrological loads and exacerbate large-scale flood risks. For instance, in 2019, Typhoon Hagibis caused record-breaking precipitation in the Kanto region of Japan, leading to the overtopping of multiple rivers and extensive urban flooding in Tokyo and its surrounding areas (Moya et al. 2020). Overall, the differences in flood causes among various urban agglomerations stem from the complex interplay of hydrological processes, where climate systems, topographical features, and human activities jointly shape flood-triggering mechanisms and evolution pathways.

4.2 Quantification and assessment of flood impacts

4.2.1 Urban scale: quantitative research on vulnerability of disaster-bearing entities

The core of flood assessment lies in accurately quantifying the vulnerability characteristics of disaster-bearing entities to reveal their risk exposure across different spatial and temporal scales (Fig. 5, second row, left). Current flood simulation methods are based on multi-scale dynamic modeling and data integration technologies. By tracking the spatiotemporal evolution of floods, these methods analyze their dynamic patterns to provide a foundation for urban disaster prevention planning (Laino et al. 2024; Saikia et al. 2022). However, these methods often focus on short-term pre-disaster predictions and in-disaster responses, with limited attention to long-term post-disaster recovery modeling. For instance, risks triggered by upstream rainstorm events are often transmitted downstream through drainage pathways (Xinyu and Qiao 2019). However, current models inadequately quantify such multi-stage risk chains, exacerbating the spread of localized risks to a broader area when drainage node capacities are insufficient (Deopa et al. 2024).

Existing studies typically limit their assessment dimensions to urban engineering systems, supply systems (traffic flow, rescue capacity), and operational systems (economic, social, environmental) (Table 1). These studies quantify the dynamic interactions between floods and various systems to reveal flood propagation patterns and impacts but remain constrained in addressing cross-regional linkages and complex feedback mechanisms in functional areas (Liang et al. 2024). From the distribution of indicators in Table 1, key indicators such as drainage capacity, storm runoff coefficient, and flood peak flow overly emphasize the coupling relationship between water and urban physical spaces, neglecting the physical and logical interconnections of engineering systems and their influence on disaster impacts (Tang et al. 2024). Meanwhile, the coverage of socio-economic indicators such as population, social vulnerability index, and poverty rate underscores the importance of resource allocation equity as a key factor in addressing flood disasters (Ma et al. 2023). Ecological and environmental indicators are critical reflections of urban sustainability, but their quantification standards are challenging to unify, and their methodologies and data acquisition are difficult, resulting in relatively low adoption in the literature(H. Li et al. 2024a, b). Moreover, while high-resolution remote sensing and machine learning significantly enhance assessment accuracy, achieving effective risk quantification under constrained budgets and incomplete data remains a core challenge.

4.2.2 Urban agglomeration scale: comprehensive quantitative research from a systems theory perspective

The more complex socio-economic structure of urban agglomerations makes them exhibit significant cross-regional coupling and multi-scale risk diffusion when facing floods. The word cloud on the left side of the third row in Fig. 5 indicates that existing research on urban agglomerations focuses on three scientific questions: (1) Evaluation of spatial heterogeneity in urban agglomerations; (2) Dynamic spatiotemporal interactions between disasters and spaces and their multi-stage chain characteristics; (3) Optimization of existing assessment frameworks from the perspective of collaboration within urban agglomerations.

Earlier studies on the spatial heterogeneity of urban agglomeration flooding often analyzed cities as independent entities, lacking exploration of the systemic interactions within the overall urban agglomeration (Zhou et al. 2022). Using a unified indicator system to individually evaluate cities within an urban agglomeration—such as analyzing each city's drainage capacity, storm exposure, and ecological functions—this node-focused comparative approach overlooks the impact of resource flows and risk diffusion pathways between cities (Wang et al. 2023a, b, c, d, e). The field has recognized that the lack of research on cross-city interaction mechanisms hinders a comprehensive understanding of the complex dynamics and linkage effects of urban agglomerations as a whole during floods. Therefore, overcoming the limitations of single-node analysis and quantifying dynamic coupling relationships across the entire network has become a primary direction for assessing flood resilience in urban agglomerations (Ji et al. 2024).

Current quantitative research on the complex interactions of urban agglomeration flood chains often relies on watershed models (e.g., SWAT), focusing on natural topography

Table 1 Dimensions, indicators, and their key roles in urban flood assessment

Evaluation dimension	Core indicators	Literature coverage	Impact type	Key role in flood impact process
Hydro-meteorological	Rainfall intensity	80%	−	Determines water volume changes and flood formation
	Runoff coefficient	60%	−	Reflects surface permeability; impervious surfaces exacerbate flooding
	Peak flow	75%	−	Determines flood propagation speed and peak intensity
	Rainfall duration	55%	−	Prolonged rainfall leads to surface saturation and increased water accumulation
	Spatial distribution of rainfall	65%	−	Uneven rainfall increases localized drainage pressure
Socio-economic	Exposed population	85%	−	Determines disaster exposure and potential damage scale
	Economic loss	60%	−	Measures the economic impact of floods
	Density of critical infrastructure	60%	+	Affects urban functionality maintenance during disasters
	Social vulnerability index	65%	−	Reflects the susceptibility and recovery capacity of vulnerable groups
	Poverty rate	40%	−	Higher poverty correlates with lower resilience and higher disaster impact
Infrastructure	Drainage network capacity	70%	+	Determines drainage efficiency, insufficient capacity causes water-logging
	Pumping efficiency	50%	+	Directly affects drainage speed and water stagnation duration
	Road network connectivity	55%	+	Influences transportation resilience and emergency response
	Resilience of water supply systems	60%	+	Ensures water supply stability post-disaster
	Subway flood resistance	40%	+	Affects underground transport vulnerability; poor resistance leads to shutdowns
Spatial Distribution	Land use patterns	65%	−	Influences runoff processes; urban expansion reduces flood retention areas
	Building density	40%	−	Constrains water diffusion, reducing drainage efficiency
	Road network distribution	55%	+	Determines flood propagation pathways and transport recovery
	Flood zoning accuracy	50%	+	Affects high-risk area identification and flood planning
	Green space ratio	60%	+	Enhances stormwater retention and reduces runoff

Table 1 (continued)

Evaluation dimension	Core indicators	Literature coverage	Impact type	Key role in flood impact process
Ecological Environment	Green coverage ratio	55%	+	Improves rainfall interception through green infrastructure
	Biodiversity index	40%	+	Indicates ecological stability and natural flood resilience
	Forest retention capacity	40%	+	Increases water interception and reduces runoff
	Wetland flood retention capacity	50%	+	Wetlands act as natural retention areas, slowing floodwaters
	Natural recovery capacity of river channels	45%	+	Determines flood duration and river system stability

and water system distribution for disaster simulation. However, these models lack clear descriptions for quantifying the dynamic linkages between urban agglomeration infrastructure and functional areas (Xu et al. 2024). Moreover, in the temporal dimension, existing studies tend to focus on single-stage risk exposure assessment (Wang et al. 2024a, b, c, d, e, f), with limited exploration of the dynamic connections between pre-disaster warnings, in-disaster responses, and post-disaster recovery. This limits their ability to capture the full-chain dynamics of floods and the transmission mechanisms within complex socio-economic contexts (Wang et al., 2024a).

Compared to the assessment framework of individual cities, urban agglomerations place greater emphasis on regional synergy and dynamic feedback mechanisms (Table 2). Physical indicators, such as drainage capacity and flood peak flow, can measure the differences in carrying capacity among individual cities. However, indicators like cross-regional flood propagation rates and upstream–downstream hydrological resource coordination remain at an isolated assessment level (Wang et al. 2024a, b, c, d, e, f). Socio-economic indicators, such as exposed population and social vulnerability indices, reflect the adaptive capacity and resource allocation efficiency of different cities (Zhu et al. 2021). However, existing studies lack sufficient analysis of these indicators within the overall framework of urban agglomerations. Furthermore, ecological indicators, such as wetland flood retention capacity and ecological buffer zone coverage, are widely recognized in theory but face challenges like data scarcity and insufficient spatial resolution in practical assessments.

4.2.3 Multi-scale flood indicator transmission mechanism

In flood assessment systems, flood mechanisms at different scales exhibit nested transmission characteristics, with corresponding indicator systems reflecting distinct focal points (Fig. 6). Moreover, within the same scale, various indicators interact, forming complex feedback mechanisms.

During the watershed–urban agglomeration–city transmission process, flood indicators exhibit significant scale dependency. At the watershed scale (macro-scale), precipitation-runoff response and watershed hydrological cycles determine the spatial distribution of interregional water flows and the potential for water resource regulation, providing the hydrological background for downstream areas (J. Wang et al. 2023a, b, c, d, e).. At the urban agglomeration scale (meso-scale), factors such as precipitation distribution and interregional water regulation patterns dominate regional hydrological control capacity, directly influencing flood pressure on downstream cities. Finally, at the city scale (micro-scale), flood risks manifest based on underlying surface characteristics and drainage system layouts, determining the severity of urban inundation. Therefore, cross-scale flood risk assessment should focus on the interconnected relationships among different hierarchical indicators to enhance the adaptability of flood prevention strategies.

Additionally, within the same scale, indicators exhibit strong interrelationships. For example, at the urban agglomeration scale, intercity flood regulation capacity depends not only on the drainage efficiency of individual cities but also on regional water resource allocation and land-use patterns (Lu et al. 2023). Precipitation distribution and hydrological connectivity determine which cities face greater flood discharge demands, while urban expansion and green space ratios, in turn, influence the overall flood regulation capacity of the urban agglomeration. At the city scale, the proportion of permeable surfaces, road drainage networks, and underground drainage capacity collectively determine how precipitation accumulates and disperses. Thus, constructing a scientifically sound indicator system

Table 2 Dimensions, Indicators, and Their Key Roles in Urban Agglomeration Flood Assessment

Evaluation dimension	Core indicators	Literature coverage	Impact type	Key role in flood impact process
Natural dimension	Coupling of interregional flood transmission paths	75%	–	Describes how floodwaters propagate between regions, influencing upstream–downstream flood interactions
	Upstream–downstream water resource regulation index	70%	+	Measures the efficiency of inter-city water management and flood mitigation
	Buffer zone retention capacity	65%	+	Determines the ability of buffer zones (e.g., wetlands, floodplains) to absorb floodwaters
	Variation in rainfall intensity across basins	60%	–	Highlights differences in rainfall distribution, affecting flood risks in different urban areas
	Linkages between natural disaster chains	55%	–	Identifies how floods interact with other hazards like landslides and storm surges
Socioeconomic dimension	Interregional economic resilience	80%	+	Assesses the ability of interconnected economies to withstand and recover from flood events
	Regional transportation Network recovery index	70%	+	Evaluates transport system robustness and recovery efficiency post-flood
	Logistics node exposure index	65%	–	Indicates flood risk to key logistics hubs, affecting supply chain resilience
	Vulnerable population resilience index	60%	+	Reflects the adaptability and recovery capacity of socially vulnerable groups
	Urbanization-driven risk exposure	50%	–	Highlights flood risks caused by rapid and unregulated urban expansion
Technological dimension	Multi-regional applicability of catastrophe models	75%	+	Supports large-scale flood risk simulations and scenario analysis
	Multi-source dynamic Data integration capability	70%	+	Enhances inter-city flood monitoring through data fusion
	Smart flood monitoring coverage	65%	+	Improves real-time flood detection across multiple cities
	Flexibility of multi-scenario simulation parameters	60%	+	Enables adaptive flood scenario planning
	Precision of remote Sensing evaluation	50%	+	Improves flood risk mapping and assessment using satellite data

Table 2 (continued)

Evaluation dimension	Core indicators	Literature coverage	Impact type	Key role in flood impact process
Ecological dimension	Connectivity of green infrastructure networks	■■■■■ 55%	+	Enhances flood absorption capacity through ecological corridors and green spaces
	Wetland ecosystem service intensity	■■■■■ 50%	+	Measures wetlands' ability to retain floodwaters and reduce urban runoff
	Forest coverage and water retention capacity	■■■■■ 45%	+	Supports flood mitigation through improved water infiltration
	Flood prevention efficiency in ecologically sensitive areas	■■■■ 40%	+	Evaluates how protected natural areas contribute to flood resilience
	Post-Disaster biodiversity recovery potential	■■■■ 40%	+	Assesses ecological regeneration capacity post-flood events

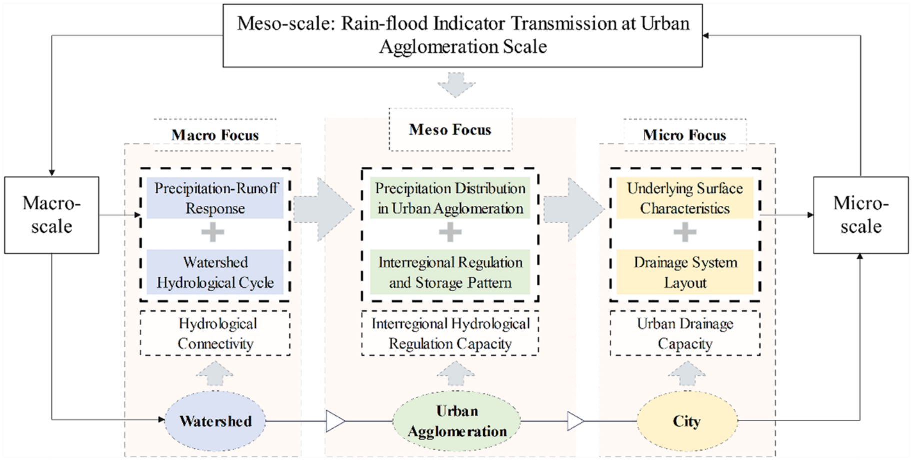

Fig. 6 Multi-scale flood assessment focus and core indicator transmission process

for urban agglomerations and cities, while uncovering their intrinsic coupling mechanisms, is crucial for improving flood risk identification accuracy and optimizing flood management decision-making.

4.3 Resilience enhancement strategies

4.3.1 Urban scale: infrastructure, social resilience, and smart governance

The word cloud on the left side of the third row in Fig. 5 indicates that existing research has gradually shifted toward a comprehensive governance approach centered on systemic resilience, employing multi-dimensional and multi-level solutions. Figure 7 illustrates that this transition reflects a shift in governance focus from traditional engineering controls to more resilient and adaptive comprehensive strategies (Asibey et al. 2022).

Green and gray infrastructure have long been considered two poles in flood management: green infrastructure demonstrates significant effectiveness in mitigating flood impacts through wetland restoration and the establishment of ecological buffer zones (Meng et al. 2022). However, its deployment in high-density urban areas is constrained

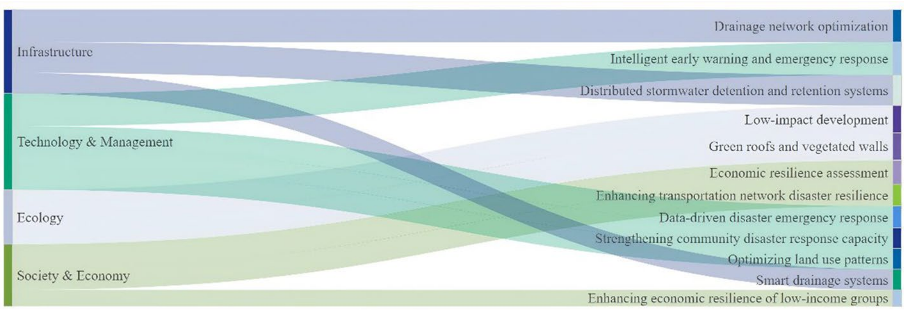

Fig. 7 Dimensions and strategies for enhancing urban flood resilience

by land resource allocation (Wu et al. 2022), a challenge that becomes more pronounced in policy environments lacking cross-sectoral coordination. Gray infrastructure, such as drainage networks and pumping stations, provides rigid protection during high-intensity rainfall scenarios However, systemic updates are challenging, and integration with smart technologies to achieve dynamic optimization and real-time adaptation at a broader scale remains difficult. Moreover, current modeling methods for the collaborative flood management of green and gray infrastructure are overly idealized. Efforts are underway to explore dynamic responsiveness to actual urban resource distribution and functional area needs, thereby enhancing the practicality of collaborative optimization strategies in resource-constrained scenarios (Li et al. 2024a, b).

Beyond strengthening physical defense systems, enhancing urban flood resilience also relies on the adaptive responses of social systems to disasters. In the current framework, social resilience is positioned as a simple combination of community participation, disaster education, and shelter networks (Hettiarachchi et al. 2022), but its inherent complexity is often overlooked. Socially vulnerable groups disproportionately bear the burden of flood vulnerability due to unequal resource distribution and limited access to information, a phenomenon especially evident in less developed regions such as India. One solution is to establish a cyclic mechanism: communities act as governance entities, embedding their capabilities into urban infrastructure development and emergency strategy design through improved risk perception and a closed feedback loop of demand (Hassani et al. 2024). The construction of this mechanism relies on policy-level support and institutional guarantees.

From a governance perspective, intelligent governance introduces data-driven dynamic decision-making capabilities for flood risk management. Research focus has shifted beyond static planning to implementation control and multi-scenario simulation. Existing studies tend to focus on technological tools themselves while overlooking the social adaptability of their applications. For example, the combination of IoT sensors and machine learning algorithms provides precise flood predictions, but their application in cross-sectoral and cross-regional collaboration still faces issues such as data silos and unequal resource distribution. This is particularly challenging for small and medium-sized cities, where technology costs further exacerbate resilience-building disparities (Guan et al. 2024; Yosri et al. 2024). Additionally, much research focuses on utilizing real-time data to optimize the dynamic layout of flood facilities while precisely quantifying community needs to enhance the full-chain capabilities of forecasting, early warning, simulation, and planning (Chen et al. 2024a, b, c, d). Floods are characterized by short durations but high intensities. In practice, the true breakthrough in systematic flood management lies in integrating the complex interactions of infrastructure, social, and management dimensions to construct an adaptive comprehensive governance framework.

4.3.2 Urban agglomeration scale: predictive models and regional collaborative governance

The dynamic optimization of disaster propagation pathways, the construction of cross-regional collaborative mechanisms, and the systematic development of disaster-adaptive networks collectively form the core framework for flood management in urban agglomerations (Fig. 5, third row, right).

Urban agglomeration governance significantly increases in complexity compared to flood management for single cities, with its core rooted in the interconnectedness of

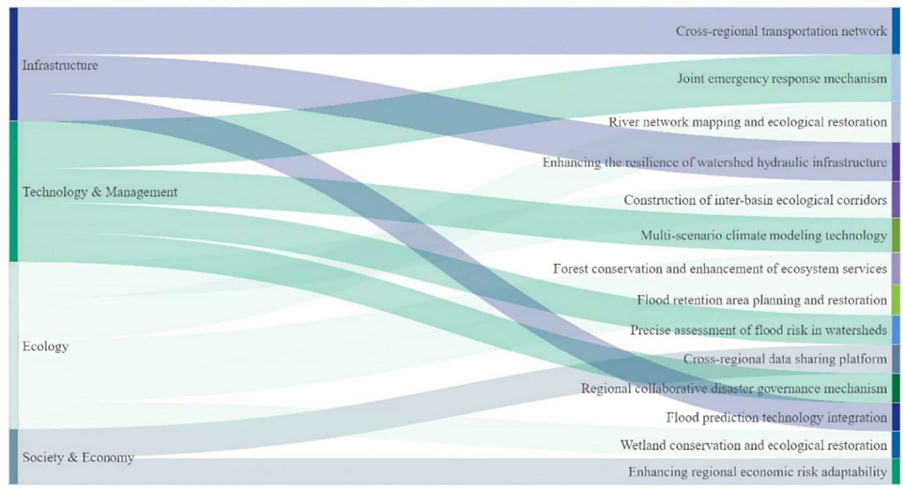

Fig. 8 Dimensions and strategies for enhancing urban agglomeration flood resilience

multi-city systems and the nonlinear characteristics of cross-regional risk propagation (Fig. 8). The complexity of propagation pathways for extreme weather events, such as typhoons and heavy rainstorms, at the urban agglomeration scale involves cascading disaster chain effects that not only impact drainage and flood defense systems within individual cities but also extend across critical infrastructure like transportation networks and water supply systems (Sun et al. 2024). There has been some progress in multi-scale cascading risk identification for disasters. For instance, dynamic simulation models based on remote sensing and big data analysis can accurately predict multi-tiered flood propagation pathways triggered by typhoons and heavy rainstorms, providing real-time decision support for risk control (Sun et al. 2024). However, challenges remain in applying IoT technologies at critical nodes along disaster propagation pathways. Additionally, cross-regional watershed flood risk management systems have gradually become a focus in governance practices. Their core lies in achieving efficient resource allocation and equitable risk sharing through regional collaboration and division of labor. For example, upstream and downstream cities' coordinated management during typhoons and floods can dynamically balance flood dam regulation and retention area functions (Hu et al. 2022). Such collaborative governance requires not only technological support but also policy consistency and equitable resource distribution.

The construction of disaster-adaptive networks for urban agglomerations represents a novel approach to enhancing regional resilience, focusing on the coordinated integration of ecological, social, and infrastructure dimensions. In recent years, research has shifted from optimizing individual disaster prevention facilities to systematically building overall disaster-adaptive capabilities (Das et al. 2024). For example, dynamic scenario-based multifunctional zone models optimize resource allocation efficiency between high-risk areas and ecological buffer zones, enhancing overall regional resilience. Existing studies have developed preliminary zoning models based on geographical and socio-economic vulnerability characteristics (Soto-Montes-de-Oca et al. 2023). However, integrating real-time monitoring data with these risk zoning models to achieve dynamic optimization and immediate response remains a critical challenge in the field.

5 Discussion

5.1 Challenges and opportunities in urban flood resilience research

The most commonly mentioned limitations of urban flood resilience include adaptability to dynamic risk propagation, the vulnerability of disadvantaged groups, and the application of intelligent technologies. However, this study argues that it is also necessary to analyze the feasibility and necessity of the current research directions in the field.

Existing studies on dynamic risk propagation and resource linkage methods rely on static zoning models for localized risk analysis, but scholars have proposed designing models that account for the complex interaction dynamics and nonlinear mechanisms of urban functional zones (Aroca-Jiménez et al. 2023). On the one hand, they aim to address the slow responsiveness of current models to cross-regional resource scheduling and disaster impact chains. On the other hand, they seek to move beyond single scenarios, incorporating the uncertainties of disaster environments and the dynamic characteristics of multi-hazard scenarios (Ding and Wu 2023). Such research is indeed necessary, as tracking regional resource flows and risk diffusion during floods helps optimize disaster relief resource allocation, provides decision-making support for cross-jurisdictional disaster collaboration, and enhances the timeliness of disaster response (Chen et al. 2024a, b, c, d).

However, the field needs to reflect on whether cities truly require real-time risk propagation models during disasters. The coupling relationship between cities and disasters is far more complex than the disasters themselves. Real-time risk propagation models consume significant computational resources and data support (Ding et al. 2024), but their real-time accuracy may fall short in rapidly changing disaster scenarios. Moreover, the actual value of such models may be overestimated in some cases, particularly in small and medium-sized cities, where the investment in financial and technical resources often outweighs the potential benefits. Super cities (e.g., New York, Shanghai) may have the financial and technological capacity to implement such high-precision models, but whether ordinary cities blindly emulate them for political achievements—leading to resource wastage and exacerbated regional inequalities—warrants further investigation. Such issues are particularly likely to arise in developing countries with insufficient governance capacity. For example, China is currently vigorously developing digital twin systems, but many regions face problems of over-hyped promotion, mismatches between technology and actual needs, and a lack of standards (Macatulad and Biljecki 2024).

In contrast, disaster forecasting models and disaster planning designs may be more practical.The advantage of these models lies in their ability to provide clear and timely warning information in a cost-effective manner before disasters occur, giving managers valuable decision-making time (Abdrabo et al. 2023). Additionally, disaster planning emphasizes not only interrupting the disaster propagation chain but also the precision and operability of resource allocation, thereby reducing the impact of disasters on core functional areas (Chang et al. 2024). For example, through dynamic assessments of potentially affected areas in advance, emergency evacuation routes and resource allocation can be optimized to ensure flexibility and specificity in disaster responses (Coletta et al. 2024). This strategy has higher general applicability and can be effectively promoted in cities with varying economic levels, avoiding resource waste and inequity potentially caused by high-cost technological tools.

The issue of social vulnerability during floods is another major limitation of current research.Low-income groups and aging communities face structural inequalities in resource allocation for disaster management, exacerbating the risk exposure of vulnerable groups (Salhi et al. 2024). The challenge lies in quantifying community feedback and incorporating it into the city's overall flood management strategy to ensure effective responses for vulnerable groups during disasters (Hart et al. 2024). Some studies propose designing idealized models to precisely calculate demand and resource allocation, but scholars should also consider the practical barriers to implementing theoretically perfect models. The core issue may lie in the government's willingness to adopt technology and whether it can bear the economic costs for vulnerable groups. It also remains questionable whether these models can accurately capture the needs of residents and vulnerable groups during disasters. In many developing countries, data scarcity and collection difficulties further hinder the applicability of these models.

5.2 Challenges and opportunities in urban agglomeration flood resilience research

Flood management in urban agglomerations, due to its cross-regional connectivity and systemic complexity, has greater heterogeneity and coordination demands compared to single cities. As discussed in the methodology section, the volume of research on flood resilience in urban agglomerations is significantly smaller than that for individual cities. The challenges analyzed in the conclusion section are mainly focused on cross-regional risk propagation and policy coordination.

As mentioned earlier, many assessments of flood resilience in urban agglomerations actually focus on measuring and comparing the resilience of individual cities within the system. The advantage of such studies lies in their ability to provide preliminary guidance for functional zoning and differentiated flood strategies within a region, particularly in urban agglomerations where geographic and socio-economic differences are not significant. These findings offer practical references for local policymaking (Zhu et al. 2024a, b). However, the core issue lies in the generalized application of indicator systems. Unified evaluation frameworks attempt to compare the risks and resilience of cities horizontally, but this approach overlooks the dynamic interaction characteristics of urban agglomerations and weakens the ability to identify cross-regional resource linkages and risk diffusion pathways. The most critical concern is the generalized use of such studies in urban agglomerations with significant economic and resource disparities between cities. The weighting of indicators and the focus of disaster prevention needs often vary with the development level and functional positioning of cities, directly leading to a lack of practical applicability in the evaluation results. A one-size-fits-all evaluation framework neither accurately measures the systemic resilience of economically imbalanced urban agglomerations nor scientifically compares and optimizes the individual resilience of each city. Future research should, on the one hand, focus on systemic resilience and construct scientific evaluation frameworks that integrate dynamic resource flows and coordination mechanisms between cities (Chen et al. 2024a, b, c, d). On the other hand, it should maintain horizontal comparative studies of individual city resilience but must design flexible frameworks capable of dynamically adjusting indicator weights to scientifically evaluate the differences in flood resilience among cities within an urban agglomeration.

Floods at the urban agglomeration scale often exhibit disaster chain characteristics involving typhoons, heavy rainstorms, and watershed flooding (Zhao et al. 2024a, b). This multi-hazard compounding effect makes upstream–downstream collaborative governance a critical challenge (Wang et al. 2023a, b, c, d, e). The dynamic propagation pathways of disaster chains are complex, involving resource allocation and interest coordination among different cities. Existing governance systems often struggle to address these issues effectively (Wang and Hou 2023). The planning and utilization of flood detention areas are often postponed due to local interest conflicts, reflecting deep-seated challenges in harmonizing regional governance interests. Existing studies consider ecological compensation mechanisms as important tools for resolving these conflicts. For example, establishing financial incentives and water resource trading systems between upstream and downstream cities can help achieve a balance of interests (Yu et al. 2022). However, in practice, disagreements over compensation standards among local governments and the lack of a mandatory policy framework often reduce collaboration to mere formalities.

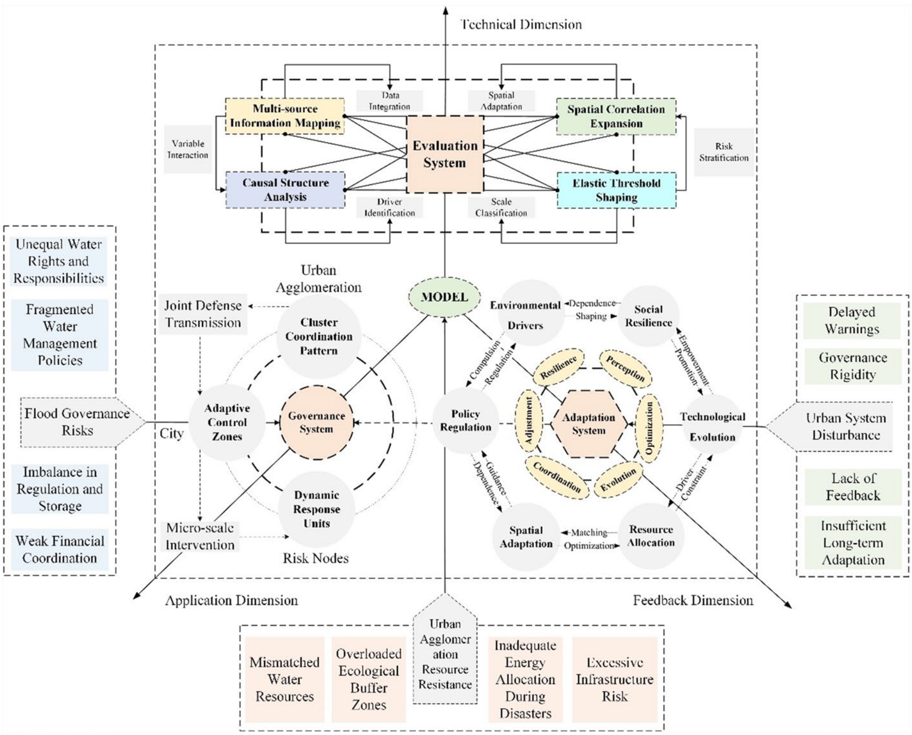

Fig. 9 Technical-policy application framework for urban agglomeration and city flood governance

Therefore, in the intercity flood management system, to avoid the shortcomings of the traditional administrative division-based governance model, a more efficient technology-policy coordination framework is needed to optimize flood response strategies. The establishment of an intelligent flood early warning system, integrating multi-source hydrological data, big data analysis, and hydrological response node detection, enables dynamic monitoring of upstream rainfall and downstream flooding (Wan et al. 2022). Through a hierarchical warning mechanism, it enhances the ability of cities within a region to respond to extreme precipitation events. Moreover, flood risk simulation and data sharing from a systemic perspective of urban agglomerations are key to improving overall adaptive capacity. This approach can break down information silos and fragmented flood control strategies caused by independent governance models, allowing for precise identification of intercity hydrological risk areas. Specifically, by integrating hydrological models, urban flood models, and historical disaster data, a comprehensive simulation platform can be developed to identify critical flood pathways and risk nodes between cities, thereby enhancing overall coordination based on shared data. Additionally, in terms of regional floodwater resource regulation, an integrated perspective should be adopted to optimize water resource allocation and flood discharge strategies across upstream, midstream, and downstream cities, preventing individual cities with insufficient disaster-bearing capacity from being placed in a passive situation. Fiscal incentives and water resource trading mechanisms can provide economic drivers for coordinated upstream–downstream flood management, thereby enhancing overall regional collaboration. The establishment of this policy-technology integration framework will help urban agglomerations improve their collective response to extreme weather events and achieve more precise flood risk management (Fig. 9).

6 Conclusions

Research on urban and urban agglomeration flood resilience is a critical field for addressing the challenges of extreme weather. Governance in individual cities focuses on managing inundation risks and optimizing resource allocation, while the complexity and diversity of urban agglomerations demand higher standards for regional collaborative governance and dynamic risk management. Existing research in the field rarely offers systematic comparative reviews or reflects on the starting points of current studies. This study conducted a mixed quantitative and qualitative analysis of 412 papers related to urban and urban agglomeration flood resilience. It is the first to use large language models and machine learning models to perform a structured review in the field of urban resilience, offering an efficient approach to data analysis and synthesis for future research on disaster prevention and mitigation in built environments.

Through comprehensive review and comparative analysis, this study provides data-supported in-depth conclusions for the field of flood resilience research and presents four findings: (1) Research on flood resilience at the urban scale significantly outweighs that on urban agglomerations, failing to systematically reveal the challenges urban agglomerations face during floods. (2) Existing studies emphasize the natural causes of floods but insufficiently explore socio-economic mechanisms and cross-regional interactions. (3) Existing indicator systems fail to scientifically evaluate resource imbalances and demand differences within urban agglomerations, making it difficult to measure systemic resilience and individual city differences effectively. (4) Although many studies propose strategies based on systemic resilience frameworks, they lack comprehensive discussion on dynamic resource allocation and policy integration mechanisms for cross-regional disaster response.

In addition, this study identified misconceptions in the field and proposed forward-looking optimization directions. First, current research overly emphasizes idealized real-time risk propagation models, neglecting the practicality and cost-effectiveness of disaster forecasting models and disaster planning designs. Moreover, while there is considerable focus on the vulnerability of low-income regions and populations, there is a lack of consideration for the practical implementation of such measures. Additionally, the weak execution of cross-regional policy coordination and ecological compensation mechanisms, along with inter-city interest conflicts, hinders the practical effectiveness of collaborative governance. Future research should focus on the implementability of cross-regional governance solutions and consider the economic feasibility and general applicability of resilience enhancement strategies.

Despite the significant innovations in the analytical methods of this study, which provide valuable insights into automated literature review and quantitative analysis, large language models still have room for improvement in capturing domain-specific details, particularly in recognizing nuanced differences in practical applications and research contexts. However, the systematic review data compiled in this study remains highly valuable for flood risk management. By leveraging the structured data from this study, cities can systematically identify key factors influencing flood risks, pinpoint high-risk areas, and optimize flood prevention planning and coordination at a regional scale. Additionally, these data can support inter-city disaster response efforts by providing quantitative insights that facilitate the development of more scientific and systematic collaborative disaster management policies. This approach helps mitigate the fragmentation and localized inefficiencies of conventional governance models, ultimately enhancing the overall flood resilience of urban agglomerations.

Appendix

See Figs. 10, 11 and 12.

```
"""
Below is a list of structured questions designed to systematically extract
information from flood resilience studies at urban and urban
agglomeration scales. The questions were formulated based on key
research dimensions, including research scale, methodology, data types,
and resilience strategies.

The purpose of this structured questionnaire is to ensure
comprehensive coverage of urban flood resilience research, enabling
large-scale automated literature analysis. Each question is formatted in
a JSON object and follows a multiple-choice or structured response
approach. The provided example answer demonstrates how responses
should be formatted.

- Answer DOI and title of the paper in a JSON format.
- Summarize the study in 3 bullet points in a JSON format based on
introduction, conclusion, and abstract. 3 points should cover:
Purpose/aim of the study, method, and findings.
- Select the research scale of the study: Urban scale, Urban
agglomeration scale, Watershed, Community level, Cross-scale, or
County level.
- Choose the research type: Quantitative research, Qualitative research,
or Mixed methods.
- Identify the analytical tools used: Statistical analysis, Simulation
modeling, Hydrological modeling, GIS analysis, Network analysis,
Machine learning, Indicator-based assessment.
- Specify the data types used: Hydro-meteorological data,
Environmental data, Social/Economic data, Emergency response data,
Infrastructure data, Remote sensing imagery.
- Determine the primary research direction: Causes of urban flooding,
Causes of urban agglomeration flooding, Flood impact quantification,
Flood resilience enhancement strategies.
- Identify which aspect of flood causes is analyzed: Disaster-bearing
system vulnerability, Human interference, Disaster chain effects.
- Indicate the quantification method for flood impacts: Indicator
scoring, Affected area quantification, Environmental impact assessment,
Economic loss estimation, Casualty statistics.
- List the flood resilience enhancement strategies discussed:
Management measures, Infrastructure improvements, Ecological
restoration strategies, Integrated governance strategies.

The structured responses ensure consistency and enable automated
analysis of resilience research across multiple scales.
"""

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        "County scale",
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        "Quantification and assessment of urban agglomeration flood
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Remote sensing imagery",
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    "Focus on flood causes": "Disaster chain effects",
    "Quantitative methods for flood impacts": "Affected area
quantification",
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governance strategies"
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}
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Fig. 10 Structured questioning mechanism

```
import os
import openai
import pandas as pd

def load_text(file_path):
    """Load the text content of a file."""
    with open(file_path, 'r', encoding='utf-8') as file:
        return file.read()

def ask_gpt_with_weighted_answers(document_text, question_prompt):
    """
    Ask GPT-4 for the answer to a question based on a document's text.
    """
    try:
        response = openai.ChatCompletion.create(
            model="gpt-4",
            messages=[
                {"role": "system", "content": "You are a helpful assistant
specialized in document classification."},
                {"role": "user", "content": question_prompt},
            ],
            temperature=0,
            request_timeout=120
        )
        answer = response['choices'][0]['message']['content'].strip()
        return answer
    except Exception as e:
        return f"Error: {str(e)}"

def process_files_with_weights(directory_path, question_prompt):
    """
    Process all text files in the directory to classify answers with weights.
    """
    results = {}
    file_list = sorted([f for f in os.listdir(directory_path) if
f.endswith('.txt')])

    for file_name in file_list:
        file_path = os.path.join(directory_path, file_name)
        print(f"Processing file: {file_name}")
        document_text = load_text(file_path)
        document_excerpt = document_text[:2000]  # Adjust token limits
as needed
        answer = ask_gpt_with_weighted_answers(document_excerpt,
question_prompt)
        results[file_name] = answer
        print(f"Answer: {answer}\n")
    return results

def summarize_weighted_results(results, categories, weights):
    """
    Summarize the results with weighted counts.
    """
    weighted_counts = {category: 0 for category in categories}

    for file, answer in results.items():
        for category in categories:
            if category.lower() in answer.lower():
                if "primary" in answer.lower():
                    weighted_counts[category] += weights['primary']
                elif "secondary" in answer.lower():
                    weighted_counts[category] += weights['secondary']

    return weighted_counts

def save_weighted_results_to_excel(results, summary, output_path):
    """
    Save the results and weighted summary to an Excel file.
    """
    results_df = pd.DataFrame(list(results.items()), columns=["File",
"Answer"])
    summary_df = pd.DataFrame(list(summary.items()),
columns=["Category", "Weighted Count"])

    with pd.ExcelWriter(output_path) as writer:
        results_df.to_excel(writer, sheet_name="Results", index=False)
        summary_df.to_excel(writer, sheet_name="Weighted Summary",
index=False)

def main():
    # Add your OpenAI API key here
    openai.api_key = "xxxxxxxxxxxxxxxxxxxxxxxx"

    # Directory containing text files
    directory_path = "xxxxxxxxxxxxxxxxxxxxxxxx"

    # Define question prompt
    question_prompt = (
        "Please classify the document into one or more categories based
on the following: \n"
        "- Category A\n"
        "- Category B\n"
        "- Category C\n"
        "- Other (please specify)\n"
        "For each category, specify if it is a primary or secondary focus."
    )

    # Define categories and weights
    categories = ["Category A", "Category B", "Category C", "Other
(please specify)"]
    weights = {"primary": 1.0, "secondary": 0.5}

    # Process all files and get results
    results = process_files_with_weights(directory_path,
question_prompt)

    # Summarize results with weights
    weighted_counts = summarize_weighted_results(results, categories,
weights)

    # Save results to Excel file
    output_path = "xxxxxxxxxxxxxxxxxxxxxxxxx"
    save_weighted_results_to_excel(results, weighted_counts,
output_path)

    print(f"Results saved to {output_path}")
```

Fig. 11 Answer statistics mechanism

```
import openai
import json

# Set up OpenAI API key
API_KEY = "openai_api_key"

# Define structured questions
questions = {
    "Urban vs. Urban Agglomeration Flood Resilience": {
        "question": "What are the key differences in flood resilience studies
between urban and urban agglomeration scales? Analyze the
differences based on research objectives, spatial scale, infrastructure
interconnection, and governance mechanisms.",
        "expected_response_format": "Provide a structured analysis
comparing urban and agglomeration resilience studies."
    },
    "Applicability of Flood Impact Evaluation Methods": {
        "question": "How do different flood impact evaluation methods
(e.g., hydrological models, indicator-based assessments, remote sensing
techniques) vary in their applicability at urban and agglomeration
scales? Provide a comparative analysis of their strengths and
limitations.",
        "expected_response_format": "List methods, their advantages,
disadvantages, and applicability to different research scales."
    },
    "Effectiveness of Resilience Enhancement Strategies": {
        "question": "What are the primary strategies used to enhance flood
resilience? Categorize them (e.g., engineering measures, ecological
approaches, governance policies) and assess their effectiveness at
different scales.",
        "expected_response_format": "Provide a categorized list of
resilience strategies with scale-based effectiveness evaluations."
    },
    "Regional Governance Models and Coordination Mechanisms": {
        "question": "How do different flood governance models (e.g.,
centralized governance, collaborative governance, multi-level
management) compare across different regions? Provide real-world
examples.",
        "expected_response_format": "List governance models, their
advantages, disadvantages, and examples from different regions."
    },
    "Future Research Trends": {
        "question": "What are the emerging trends in flood resilience
research? Identify current research gaps and suggest future research
directions.",
        "expected_response_format": "Summarize research gaps and
outline key trends shaping future flood resilience studies."
    }
}

# Function to query GPT-4o
def query_gpt(question):
    response = openai.ChatCompletion.create(
        model="gpt-4o",
        messages=[{"role": "system", "content": "You are an expert in flood
resilience research. Answer the following question comprehensively: " +
question}],
        api_key=API_KEY
    )
    return response["choices"][0]["message"]["content"]

# Loop through questions and retrieve responses
responses = {}
for category, details in questions.items():
    responses[category] = {
        "question": details["question"],
        "response": query_gpt(details["question"])
    }

# Store responses in JSON format
output_file = "flood_resilience_responses.json"
with open(output_file, "w", encoding="utf-8") as f:
    json.dump(responses, f, ensure_ascii=False, indent=4)

print(f"Structured responses have been saved to {output_file}")

Expected JSON Output Format
{
    "Urban vs. Urban Agglomeration Flood Resilience": {
        "question": "What are the key differences in flood resilience studies
between urban and urban agglomeration scales?",
        "response": {
            "Spatial Scale": "Urban studies focus on a single city's flood
resistance, while agglomeration studies assess multi-city interactions.",
            "Infrastructure Interconnection": "Urban resilience is limited to
municipal drainage networks, whereas agglomeration resilience
requires inter-city hydrological planning.",
            "Governance Mechanism": "Urban governance is handled by
municipal governments, while agglomeration resilience requires
cross-jurisdictional coordination."
        }
    },
    "Applicability of Flood Impact Evaluation Methods": {
        "question": "How do different flood impact evaluation methods
(e.g., hydrological models, indicator-based assessments, remote sensing
techniques) vary in their applicability at urban and agglomeration
scales?",
        "response": {
            "Hydrological Models": {
                "Advantages": "Provides precise flood propagation
simulations for urban areas.",
                "Disadvantages": "Computationally intensive for large-scale
agglomerations."
            },
            "Indicator-Based Assessments": {
                "Advantages": "Integrates socio-economic and environmental
factors into resilience analysis.",
                "Disadvantages": "Data standardization is challenging across
regions."
            },
            "Remote Sensing Techniques": {
                "Advantages": "Enables large-scale flood monitoring and
assessment.",
                "Disadvantages": "Limited resolution for real-time flood
response."
            }
```

Fig. 12 Automated structured query system for second-round comparative flood resilience analysis

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Declarations

Conflict of interest The authors declare no conflict of interest in this study.

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